

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2022

ESE 4115

(Power Plant Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by power plant engineering? Describe the classification of power plant. 12
- 1(b). Define the following terms: 06
(i) Plant use factor, (ii) Demand factor.
- 1(c). Distinguish between load curve and load duration curve. 06
- 1(d). A power station has a daily load cycle as follows: 11
300 MW for 8 hours; 150 MW for 6 hours; 130 MW for 3 hours; 100 MW for 7 hours. If the power station is equipped with 2 sets of 200 MW each, calculate (i) daily load factor, (ii) plant capacity factor, and (iii) daily requirement of fuel if the calorific value of the fuel were 10,000 kcal/kg and the average heat rate of station were 25,000 kcal/kWh.
- 2(a). What are the effects of variable load on power plant design and operation? 09
- 2(b). What factors should be considered while selecting the capacity of a power plant? 06
- 2(c). Describe the costs involved in the generation of electrical energy. 12
- 2(d). What is tariff? Make a comparison between straight line meter rate and step meter rate tariffs. 08
- 3(a). Which fuels are widely used for power generation in Bangladesh? Make a comparison of generation cost of BPDB's own power plants according to fuel type. 10
- 3(b). Explain the power demand of Bangladesh up to 2041 according to the power system master plan 2015. 08
- 3(c). What is meant by load forecasting? What factors influence the growth of load? 07
- 3(d). Why is load forecasting required? 10
- 4(a). Describe the working principle of a hydro-electric power plant with neat sketch. 10
- 4(b). Define the following terms: 09
(i) Run-off, (ii) Hydrograph, (iii) Specific speed.
- 4(c). What factors should be considered while selecting turbine for hydro-electric power plant? 05
- 4(d). For a hydro-electric power plant, the following data is available: 11
Annual rainfall: 800 mm
Catchment area: 150 sq. km
Head: 200 m
Yield factor to allow for run-off and evaporation loss: 60%
Overall efficiency: 75%
Annual load factor: 45%.
Determine: (i) Average power produced, (ii) Capacity of the plant.

SECTION – B

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|-------|--|----|
| 5(a). | Draw the typical layout of a diesel engine power plant. Explain the fuel supply system in this plant. | 11 |
| 5(b). | Write down the advantages and disadvantages of diesel power plant. | 09 |
| 5(c). | What are the necessities of cooling system in diesel power plant? Describe the thermostat cooling system with necessary figure. | 15 |
| 6(a). | Write down the applications of gas turbine power plant. | 05 |
| 6(b). | Explain the various method used for the improvement of thermal efficiency of gas turbine power plant. | 15 |
| 6(c). | Explain the combined working of gas turbine and steam power plant. | 15 |
| 7(a). | Draw a general layout of a steam power plant and discuss its various circuit. | 12 |
| 7(b). | With the help of schematic diagram, describe the working principle of binary vapor cycle power plant. | 11 |
| 7(c). | An ideal gas turbine cycle with two stages of compression and two stages of expansion has a pressure ratio of 2.83 in each stage. Air enters each stage of the compressor at 300 K and each stage of the turbine at 1300 K. Determine the back work ratio and the thermal efficiency of this gas turbine cycle assuming (i) no regenerator, and (ii) an ideal regenerator with 100% effectiveness. | 12 |
| 8(a). | What are the impurities that may be presented in natural water? What problems may be caused by these impurities in power plant? | 12 |
| 8(b). | What is meant by draught? With a neat sketch, describe the working principle of balanced draught system. | 08 |
| 8(c). | What is cooling tower? Classify it. With the help of schematic diagram, describe the working principle of induced draught counter flow wet cooling tower. | 15 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2022

ESE 4011

(Materials for Energy Engineering Application)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | What are the different types of losses encountered in solar cells? Explain the concept of optical losses in solar cells with the proper schematic diagram. | 10 |
| 1(b). | Explain the various structures of crystalline silicon materials used in solar cells. Briefly explain the different structures of a solar cell based on the diffusion and drift device concepts. | 12 |
| 1(c). | What is meant by high-efficiency concepts of solar cells? Explain with a proper schematic diagram how the PERL concept contributes to the overall advancement of crystalline silicon solar cell technology. | 13 |
| 2(a). | Mention a third generation solar cell which is belonging to the group of thin film solar cells. Explain the working principle of such solar cell with a suitable diagram. | 10 |
| 2(b). | What are the common combination of elements used to form chalcopyrite materials, and provide a brief explanation of the layer structure of a typical CdTe solar cell and its working principle. | 12 |
| 2(c). | Briefly explain the challenges associated with the multijunction solar cell? What is spectral conversion in the context of solar cells, and how does it enhance their efficiency? | 13 |
| 3(a). | What is a fuel cell and how does it differ from a conventional battery? Also briefly explain the working principle of a fuel cell. | 10 |
| 3(b). | How are the individual fuel cells connected within a fuel cell stack? Provide a brief explanation of the layer structure of a Direct Methanol Fuel Cell (DMFC). | 13 |
| 3(c). | What is the role of PTFE in alkaline fuel cells (AFCs)? What are the functions of bipolar plates in Proton Exchange Membrane Fuel Cells (PEMFCs)? | 12 |
| 4(a). | Briefly explain the construction and operating principle of molten carbonate fuel cells (MCFCs). Also what are the general features of MCFCs. | 10 |
| 4(b). | What are the properties and functions of phosphoric acid electrolytes in phosphoric acid fuel cells (PAFCs)? Provide a brief explanation of the electrode and catalyst in PAFCs. | 13 |
| 4(c). | Explain the favorable characteristics of a solid oxide fuel cell. Explain the types of electrolytes used in this fuel cell. | 12 |

SECTION – B

- 5(a). Why does electric double layer capacitor (EDLC) show high capacitance? Describe the fundamental principles of EDLC with necessary figure. 12
- 5(b). If two capacitors $C_1 = 15 \text{ F}$ and $C_2 = 8 \text{ F}$ were connected in series and parallel to a 50 volt battery, determine the voltage across each capacitor for each type of connection. 06
- 5(c). “Pseudocapacitor is a kind of supercapacitor which is different from double layer capacitor” – explain why? 07
- 5(d). Describe the hydrogen adsorption mechanism in carbonaceous materials. What are the key parameters for increasing H_2 storage capacity in CNTs? 10
- 6(a). How do carbonaceous material such as carbon nanotubes contribute to hydrogen storage and what are their potential application in this field? 10
- 6(b). Define sputtering. Describe the sputtering process with suitable figure. 09
- 6(c). Discuss the process of chemical vapor deposition (CVD) and its role in the fabrication of thin films, powders, and single crystals? 10
- 6(d). Compare and contrast between PVD and CVD process. 06
- 7(a). Explain how will you obtain monochromatic $K\alpha$ radiation from the continuous X-rays and other characteristic X-rays. 08
- 7(b). Draw a geometric arrangement of X-ray diffractometer with functions of each component. 09
- 7(c). Crystal of cobalt (Co) is analyzed using $\text{CuK}\alpha$ ($\lambda = 1.5418\text{\AA}$) X-ray radiation. Calculate the angle of reflection, 2θ , of the (1 1 1) plane. The lattice parameter (a) of FCC cobalt is $3.544\text{\AA}$. 06
- 7(d). Calculate the structure factor for FCC unit cell and show six planes from where reflection may occur or not. 12
- 8(a). Differentiate between backscattered electrons and secondary electrons in terms of their formation and role during image generation in SEM. 06
- 8(b). How does scanning electron microscopy (SEM) enable the examination of microscopic structures with high resolution and depth of field? 12
- 8(c). How are diffraction pattern and image simultaneously present in TEM? 08
- 8(d). You need to analyze a ceramic sample (0.1 mm thick) in TEM. Explain how you will prepare the sample for TEM investigation. 09

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2022

ESE 4105

(Fuel Combustion and IC Engines)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.
iv) Table, chart may be provided on request.

SECTION – A

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|-------|--|----|
| 1(a). | Briefly explain the measurement methods and significance of the following properties:
(i) API Gravity, (ii) Reid Vapor Pressure, (iii) Aniline Point and Diesel Index | 15 |
| 1(b). | Compare and contrast ultimate analysis and proximate analysis. | 08 |
| 1(c). | Describe the possible pathways of using alcohol based fuels and hydrogen fuel for IC engines. | 12 |
| 2(a). | What are the effects of overcooling the IC engines? What are the advantages and disadvantages of different cooling systems? | 14 |
| 2(b). | What are the various components to be lubricated in an engine? Write the selection criteria of lubricating oils in various weather conditions. | 11 |
| 2(c). | Why it is advantageous to dilute fuel-air mixture? How quasi-static effect influences the volumetric efficiency of an internal combustion engine. | 10 |
| 3(a). | Using proper sketch, discuss various fuel injection systems and injector nozzles. Compare their relative merits and demerits. | 16 |
| 3(b). | What are the impacts of scavenging? Why scavenging requires more attention in case of 2-stroke engine than that for 4-stroke engine? | 10 |
| 3(c). | What is sac volume? What is the relationship between NO _x formation and combustion time? | 09 |
| 4(a). | What is smog? Discuss the mechanisms of NO _x formation? How do reduce this emission? | 12 |
| 4(b). | With proper sketch, discuss the construction and operation of a catalytic converter. | 11 |
| 4(c). | What is particulate emission? How particulate emission can be measured? Why does particulate emission from CI engines differ significantly than that of SI engines? | 12 |

SECTION – B

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|-------|--|----|
| 5(a). | Discuss the following engine performance parameters with respect to engine speed using necessary diagrams:
(i) Brake power, (ii) Brake torque, (iii) Brake specific fuel consumption. | 12 |
| 5(b). | What is meant by pumping loss? Why pumping loss is higher in conventional SI engine compared to CI engine? How to reduce it? | 13 |
| 5(c). | What is the physical meaning of SIT? Explain the key characteristics of premixed flame and diffusion flame. | 10 |

6(a).	What is meant by combustion stoichiometry? Distinguish fuel rich combustion from fuel lean combustion.	11
6(b).	"CI engine performance is smoke limited"-Justify the statement.	10
6(c).	Describe the different stages of combustion in CI engine with the help of rate of heat release (RDHR) verses crank angle (θ) diagram.	14
7(a).	What is meant by surface ignition? "Pre-ignition is more harmful than post-ignition"-Justify the statement.	11
7(b).	Why do engine inlet valves and exhaust valves open earlier and close later? Why are inlet valves made larger than exhaust valves?	14
7(c).	How knocking in SI engines differs from that in CI engines? Explain with appropriate diagrams.	10
8(a).	Discuss the limits of supercharging in both SI engine and CI engine.	14
8(b).	Why 2-stroke CI engines find wider use whereas 2-stroke SI engines are used only to a limited extent?	08
8(c).	What is meant by end gas auto-ignition? "SI engine performance is knock limited but CI engine not"- Justify the statement.	13

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2022

ESE 4007

(Smart Grid Technology)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|--|----|
| 1(a). | What are the differences between an active and a passive distribution network? Describe the operation of an active distribution management scheme with necessary block diagrams. | 15 |
| 1(b). | Who are the main stakeholders of smart grid systems? Describe their functions. | 10 |
| 1(c). | Mention the key drivers of smart grid. Is a smart grid necessary for Bangladesh? Justify your opinion. | 10 |
| 2(a). | Define the term ‘distributed generation (DG)’. What are the advantages of DG? | 11 |
| 2(b). | Describe the roles of sensing, measurement, control and automation technologies for a smart grid. | 12 |
| 2(c). | Define HAN, NAN and WAN. Compare the conventional and smart meter. Write down the application areas of the following communication technologies in smart grid: Bluetooth, WiMAX, and WLAN. | 12 |
| 3(a). | What is a smart meter? Using a diagram, show the typical architecture for communication between a smart meter and the utility server. | 09 |
| 3(b). | What is advanced metering infrastructure in smart grid (AMI)? Write down the differences between AMR and AMI. | 06 |
| 3(c). | What is meant by demand side management? Describe how the following techniques modify the load consumption patterns:
(i) Load shifting, (ii) Valley filling, and (iii) Peak clipping. | 12 |
| 3(d). | Draw the configuration of a typical microgrid and describe its operation. | 08 |
| 4(a). | What is a phasor measurement unit in smart grid? Discuss a synchronized phasor measuring system with conceptual diagram. | 10 |
| 4(b). | Make a comparison between the old and desired load flow technique in smart grids. | 10 |
| 4(c). | Write a short note on the static security assessment (SSA) of smart power systems. | 05 |
| 4(d). | What is meant by a multi-agent system (MAS)? Develop an intelligent distributed autonomous power system (IDAPS) using the MAS approach. | 10 |

SECTION – B

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|-------|---|----|
| 5(a). | How many buses are there in the power system? What is the significance of different types of buses in the power system? | 06 |
| 5(b). | What are the power quality issues with smart grid? What is the importance of power quality audit in smart grid? Discuss. | 08 |
| 5(c). | Why is power grid frequency important? How does the turbine speed governing system work? Explain with schematic diagram. | 12 |
| 5(d). | What is Automatic Generation Control (AGC) in power system? “AGC is essential in smart grid” – Justify. | 09 |
| | | |
| 6(a). | Distinguish between transient and steady-state stability and discuss the need for performing stability analyses of power system. | 08 |
| 6(b). | Elucidate the importance and role of the swing equation in power system stability analysis. | 06 |
| 6(c). | Discuss the possible solutions of the disturbed motion of a synchronous generator connected to an infinite bus, when subjected to a small disturbance represented by $\Delta\delta$ and prove that, | 14 |
| | $\omega_n = \sqrt{\frac{\pi f_0}{H} P_s}$ | |
| 6(d). | Why does the grid need reactive power? Discuss. | 07 |
| | | |
| 7(a). | Why does voltage instability occurs in power grid? Discuss the voltage instability preventive control scheme for the power grid. | 08 |
| 7(b). | How decision support tools can be used in smart grid planning and operations? Describe. | 10 |
| 7(c). | What is the role of Analytical Hierarchical Programming (AHP) method in smart grid management? Explain with necessary block diagram. | 10 |
| 7(d). | Explain why cyber security is prime importance in smart grids and how the risk of cyber-attack can be mitigated. | 07 |
| | | |
| 8(a). | How does a smart grid improve efficiency? What are the benefits of distributed generation control in smart grid? | 08 |
| 8(b). | Give an overview of sustainable energy options for the smart grid. | 07 |
| 8(c). | Why are smart grid standards necessary? What would be the suitable areas of research in smart grid? | 08 |
| 8(d). | Describe the computational challenges faced by researchers and engineers for the design, planning and operations of smart power system. | 12 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2022

ESE 4125

(Safety and Environment Aspects of Energy Projects)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

iv) Table, chart may be provided on request.

SECTION – A

- 1(a). Define hazard. What are the typical sources of hazards and consequences of them? 10
- 1(b). Explain different categories of hazards according to the working situation. Let a safety valve of a system have a failure rate of once every two years or 0.5/year and the test interval is 1 week, and the system pressure is expected to rise once a year. Calculate the fractional dead time and hazard rate. 15
- 1(c). Discuss about the measures to control the hazards. 10
- 2(a). What are the key factors that determine the severing and consequences of electrical injuries? What are the risks posed by biological hazards in different occupational settings? 12
- 2(b). Briefly explain the fire classes based on the types of fuel involved, and provide an overview of commonly used fire-fighting materials for each class. 13
- 2(c). What is BLEVE? “All combustible materials do not creat explosion”- explian. 10
- 3(a). How personal protective equipment (PPE) is used in general industry to protect against various hazards associated with humans? 10
- 3(b). What are the major cause of workplace accidents? Explain the accident that occurs due to imbalance of Man and Machine relationship. Suggest the remedies of this type of accident. 13
- 3(c). Brifely explain the term accident investigation and reporting. Also, prepare a simple accident reporting sheet. 12
- 4(a). Make a list of various kinds of hazard identification methodologies. Describe the anatomy of a process incident mentioning their different modes, states, and goals. 12
- 4(b). Consider a design concept that feeds hydrogen sulfide (H_2S) from a cylinder to a processing unit. At this stage of design, the analyst knows only that this materials will be used in the process, noting more. The analyst recognizes that H_2S has toxic and flammable properties, and the analyst indentifes the potential release of H_2S as a hazardous situation. Considering the situation do a preliminary hazard analysis(PHA). 11
- 4(c). Define and explain the followings: 12
(i) Accident frequency rate, (ii) Accident severity rate, (iii) Fatal accident rate, (iv) HAZAN, (v) HAZOP, (vi) EPP.

SECTION – B

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|---|----|
| 5(a). Write short notes on the followings:
(i) Occupational health, (ii) Occupational safety and health Act. | 06 |
| 5(b). Discuss about the differences between occupational diseases and normal diseases. Write down the major causes of following occupational diseases:
(i) Lung disease, (ii) Sore eye, (iii) Blood disorders, (iv) Urticaria. | 15 |
| 5(c). Define occupational ergonomics. Explain the importance of it in workplace. | 07 |
| 5(d). Explain the necessity of study of the subject “Safety and Environmental Aspects of Energy Projects”. | 07 |
| 6(a). What does mean by occupational exposure? What are the two terms that reflects the level of exposure and differentiate between them. | 08 |
| 6(b). Write down the information that can be obtained from the followings:
(i) TLV-TWA, (ii) TLV-STEL, (iii) TLV-C. | 06 |
| 6(c). Write down the common categories of hazardous chemicals and explain their effects on health. How toxic substance affects on surroundings? | 09 |
| 6(d). Define and explain the followings with examples:
(i) Inhalation, (ii) Ingestion, (iii) Systemic effects, (iv) Chronic effects. | 12 |
| 7(a). Briefly discuss about MSDS. Explain the responsibilities to MSDS of (i) Supplier, (ii) Employer, (iii) Worker. | 12 |
| 7(b). Classify the materials that create pollution including their definition and examples. | 07 |
| 7(c). Explain the following terms of environmental impact:
(i) Local, (ii) Regional, (iii) Global. | 06 |
| 7(d). How global warming caused by the greenhouse gases? What are the measures to prevent
(i) Acid rain, and (ii) SMOG. | 10 |
| 8(a). “ Sustainable development calls for pollution prevention and clean technology”- explain. How to overcome the challenges of sustainable development? | 10 |
| 8(b). Why are IEE and EIA studies required for a project? What type of assessment is necessary for a red-category project? Explain it with the help of flow chart. | 12 |
| 8(c). Write down the procedure to get environmental clearance for a new project according to regulation Act in Bangladesh. | 06 |
| 8(d). Mention the purposes of ISO 14001. What are the major components of ISO 14001? | 07 |

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